

3. PROJECT DESIGN PHASE

Medical Diagnosis Center by Service Portal (ServiceNow)

ServiceNow IT Service Management (ITSM)

3.1 Introduction

The project design phase converts the identified requirements into a practical ServiceNow solution. The design defines the application structure, database tables, relationships, user roles, Service Portal components, and automation required for the Medical Diagnosis Center. The system is designed to provide a centralized platform for managing patients, diagnostic tests, appointments, and reports.

3.2 High-Level Design

Patient Registration → Diagnostic Test Selection → Appointment Booking → Appointment Processing → Appointment Completion → Report Generation → Report Access

For cancellation:

Appointment → Cancellation Request → Refund Calculation → Appointment Update

The system connects the Patient, Diagnostic Test, Appointment, and Report records to maintain a complete flow of information.

3.3 ServiceNow Components

Component	Purpose
Patient Table	Create and maintain patient information and registration details.
Diagnostic Test Table	Store available diagnostic tests and their relevant information.
Appointment Table	Create, update, and track patient diagnostic appointments.
Report Table	Store medical report information generated after completed appointments.
Service Portal	Provide a user-friendly interface for patients to access diagnosis center services.
Service Portal Widgets	Display forms, test information, appointment details, and reports.
Business Rules	Automate appointment cancellation, refund calculation, and report creation.

Component	Purpose
Application Scope	Organize the Medical Diagnosis Center application within ServiceNow.
Roles & Access Control	Control access to patient, appointment, test, and report information.

3.4 Database Design

The application uses separate tables to organize the major entities of the Medical Diagnosis Center.

The **Patient** table contains information about registered patients. The **Diagnostic Test** table stores the available tests and their details. The **Appointment** table connects patients with selected diagnostic tests and stores appointment-related information. The **Report** table stores the report generated for a completed appointment.

The major relationship can be represented as:

Patient → Appointment → Diagnostic Test

Appointment → Report

This structure helps maintain organized and traceable records throughout the diagnosis process.

3.5 Role Design

Role	Responsibility
Patient	Register, view diagnostic tests, book appointments, and access available report information.
Diagnosis Center Staff	Manage patient, test, appointment, and report records.
Medical/Diagnostic Staff	Process diagnostic appointments and provide required report information.
ServiceNow Administrator	Configure the application, tables, portal, security, and system settings.
Management	Monitor diagnosis center operations and review relevant records.

3.6 Related Record Design

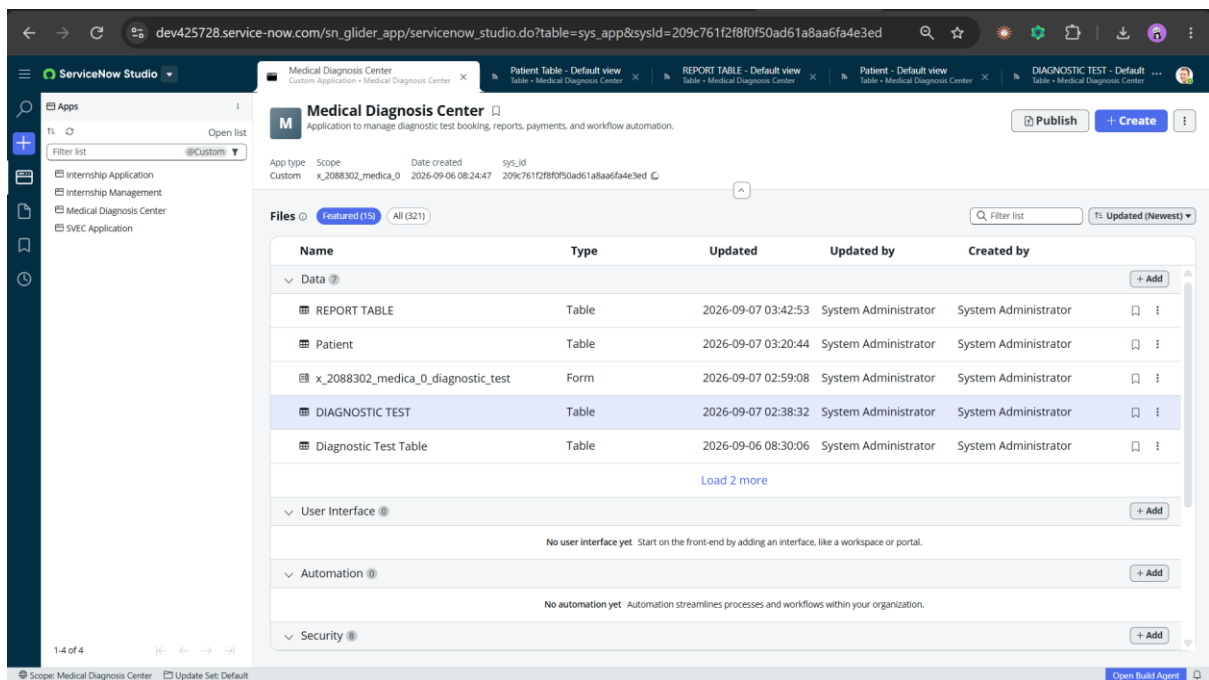
The application connects related records so that information can be tracked throughout the complete diagnosis process. A patient record is associated with the corresponding appointment, while each appointment is associated with a selected diagnostic test. After the appointment is completed, the system can create the corresponding report record.

For cancelled appointments, the appointment record is updated and the applicable refund amount is calculated according to the configured cancellation logic.

3.7 Design Result

The project design provides a structured ServiceNow application in which patient registration, diagnostic test selection, appointment booking, appointment processing, cancellation handling, and report generation are connected through related records.

The design supports centralized record management and automation while providing patients with access through the Service Portal. This approach reduces manual processing and provides better visibility into the diagnosis center workflow.



Service Portal screenshot